

UMAT-OTI workbench

Generate and verify derivatives of an Abaqus UMAT, without editing the subroutine.

1. Source
2. Material model
3. Derivatives and loading
4. Run and results

m3_j2 · fixed form · 1 routine · 0 local Newton solves · NTENS 6 · NSTATV 1 · 4 constants

2. Material model

Describe the material the subroutine expects.

Material constants

PROPS is the vector of constants the subroutine reads. Tick the ones you want derivatives with respect to; each ticked constant becomes one differentiation direction.

	≡ PROPS index	≡ Name	≡ Value	≡ Differentiate
	1	E	210000	☒
	2	nu	0.3	☒
	3	SIGY0	250	☒
	4	H	2000	☒

read from the example's contract — 4 constants, 4 marked for differentiation

State variables

STATEV holds whatever the model carries from one increment to the next, such as accumulated plastic strain.

≡ STATEV index	≡ Name
1	EQPLAS

read from the example's contract

> Advanced settings — tensor dimensions